Cluster Training: Theoretical and Practical Applications for the Strength and Conditioning Professional

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Summary

The ability to stimulate specific physiological and performance adaptations is in large part predicated by the ability to vary the training stimuli and induce novel stimuli at appropriate times. One new method for introducing training variation, while maximizing the velocity and power output of the training exercise is the utilization of a cluster set configuration. This review takes the theoretical and scientific foundation about the use of the cluster set and offers examples of how this unique training tool can be applied to the preparation of athletes from various sports.

Introduction

When examining the development of periodized training plans, the ability to introduce appropriate training variation in a logical and systematic fashion allows for the development of specific physiological and performance outcomes.[5,14] When examining the implementation of periodized strength training programs, variation can be introduced at numerous levels including manipulations of the overall training load, number of sets and repetitions, exercise order, number of exercises, focus of the training block, and the rest interval between sets.[5,20,24] Another potential level of variation that is often overlooked in contemporary strength training is the actual structure of the training set being implemented.[5,6] By manipulating the structure of the set, it is possible to induce specific adaptive responses that could favor the development of maximal strength, explosive strength, muscular growth or power endurance.

In the classic sense, a training set is comprised of a series of repetitions which are performed in a continuous fashion.[5,6] Another potential set format, which has recently been termed a cluster set[6], employs a 15–45 second period of rest between each repetition. While the main manipulation employed in the application of the cluster set involves modifications of the inter-repetition rest interval, other modifications can be used, such as varying the individual repetition loading patterns contained in the set. Generally, two global inter-repetition loading patterns can be used; the undulating and the ascending cluster set.[5] In the undulating cluster, the resistance utilized during the set is increased in a pyramid fashion[6], while the ascending cluster configuration utilizes increases in resistance with each successive repetition across the set.[5] An additional cluster set modification could be alternating the number of repetitions employed.[2,11] For example, cluster sets can be designed which contain 5-30 seconds of rest between each individual repetition, (i.e. 10/1 = 10 total repetitions in the set with inter-repetition rest between each repetition), or between a series of repetitions (i.e. 10/2 = 10 total repetitions in the set with inter-repetition rest after each series of 2 repetitions)*(Table 1)*. Ultimately, the implementation of various cluster set models should not be randomly employed, but should be constructed in the context of the overall training plan and based upon the training goal of each phase of training contained within the training plan.

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Table 1: Example Cluster Set Structures for the Power Snatch during a Strength-Endurance Phase of a Periodized Training Plan.

Type of Cluster	Sets	X	Repetitions	Example Cluster Set Repetition Loading Structure (weight / repetition)										Inter-Repetition Rest Interval (s)
Standard	1-3	X	10/1	90/1	90/1	90/1	90/1	90/1	90/1	90/1	90/1	90/1	90/1	5
	1-3	X	10/2	90/2	90/2	90/2	90/2	90/2						10
	1-3	X	10/5	90/5	90/5									15
Undulating	1-3	X	10/1	82.5/1	87.5/1	92.5/1	97.5/1	102.5/1	97.5/1	92.5/1	87.5/1	82.5/1	80/1	5
	1-3	X	10/2	85/2	90/2	95/2	90/2	85/2						10
Notes:	10/1 = 10 total repetitions broken into 10 clusters of 1; 10/2 = 10 total repetitions broken into 5 clusters of 2; 10/5 = 10 total repetitions broken into 2 clusters of 5. All weights based upon max power snatch of 120 kg (90 kg = 75% of 1 repetition maximum). Each set has an average intensity of 90kg or 75% of 1-repetition maximum). Rest intervals can be lengthened to 30 seconds depending upon the goal of the training plan and the athlete's level of development.													

Theoretical and Scientific Basis for Cluster Set Training

Conceptually, the inclusion of inter-repetition rest intervals during a set could improve the quality of the individual repetitions when compared to a traditional set structure.[5,6] When examining the traditional set configuration the barbell velocity, peak power output, and displacement would decrease across each repetition contained in the set as a result of the accumulation of fatigue.[6] The fatigue generated during this type of set structure may manifest itself as neuromuscular system fatigue or by an accumulation of metabolic fatigue inducing factors, which collectively could significantly impact performance capacity. Support for this contention can be seen in the scientific literature which reports that 5–9 maximal contractions can induce reductions in maximal force generating capacity, rate of force development, and the rate of relaxation.[23] It is likely that when performing traditional strength training sets, the use of continuous repetitions also results in a greater depletion of phosphocreatine (PCr) and a potential increased reliance on glycogen which may result in an increased production of lactate[4], which may negatively impact force generating capacity[17] when compared to cluster sets. Repetitive maximal contractions result in significant reductions in adenosine triphosphate (ATP) and PCr coupled with substantial elevations in lactate concentrations.[17] The resultant increase in lactate appears to correspond to significant reductions in force generating capacity as a result of impairments in ATP production that result in partial changes in contractile characteristics. Conversely, the addition of 15 seconds of recovery can result in a restoration of ½ of the fatigue induced force decrement returning the force generating capacity to ~79.7±2.3% of initial capacity as a result of partial replenishment of fuel substrates.[17]

The addition of short inter-repetition rest intervals would therefore offset the fatigue induced reductions in performance by allowing the athlete to partially recover between each repetition contained in the set. Ultimately this results in a higher average training stimulus across the set in response to performing the exercise with higher power outputs[11], barbell velocities[6] and barbell displacements.[5,6] Utilizing this programming strategy may be of particular benefit in the development of power as the additional rest would reduce inter-repetition fatigue and allow for higher powers to be trained with each repetition.[10,15] Based upon this evidence and logic it may be suggested that cluster set-loading paradigms are best suited for explosive or ballistic exercises such as those seen in programs that rely on weightlifting movements.[5,11]

Conversely, traditional sets may be better suited when attempting to induce hypertrophy or significantly increase muscular strength.[15] While not all studies agree[3], Rooney *et al.*[15] suggest that traditional sets produce greater strength gains even though they produce more fatigue when compared to cluster sets. It is possible that the adaptive response to traditional sets results from an increase in the activation of high threshold motor units.[11.15] Additionally, higher threshold motor units may be activated in response to the significant metabolic (i.e. lactate) induced fatigue stimulated by the traditional set configuration.[7,13,16] This increased accumulation of lactate may also produce an environment which favors hypertrophic responses.[9]

An additional consideration in the implementation of the cluster set when working with explosive or ballistic exercises is the loading paradigm employed. If undulating loading paradigms are employed, it is possible to activate a post-activation potentiation effect.[5] With an undulating set configuration, the first ½ of the set utilizes increasing loads, while the second ½ uses descending loads *(Table 1)*.[6] During the descending portion of the cluster, a potentiation effect is expressed as increases in barbell velocity, power outputs and barbell displacements.[6] It is likely that the mechanism behind these potentiation effects occurs as a result of increased phosphorylation of myosin regulatory light chains 18 or a neural effect in intact muscle 1.

Table 2: Example Cluster Set Implementation during a Hypertrophy Phase of Training

Exercise	Sets x Repetitions	Set Type	Intensity kg	Intensity % 1-RM	Inter-Repetition Rest Interval(s)
Power Clean	3 x 10/2	Standard Cluster	97.5	65	15
Back Squat	3 x 10	Traditional	162.5	65	0
Behind Neck Press	3 x 10	Traditional	75	68	0
Leg Curl	3 x 10	Traditional	60	---	0
Front Raise	3 x 10	Traditional	15	---	0
Note: Max Power Clean = 150 kg; Max Back Squat = 250 kg; Max Behind Neck Press = 110 kg					

Table 3: Example Cluster Set Implementation during a Hypertrophy Phase of Training

Exercise	Sets x Repetitions	Set Type	Intensity kg	Intensity % 1-RM	Inter-Repetition Rest Interval(s)
Power Snatch Warm-up	3 x 10	Traditional	40	29%	0
Snatch Grip Shrug	3 x 10	Traditional	126	90%	0
Rest 10 minutes					
Snatch	1 x 10/1	Undulating	91	65%	30
Snatch Grip RDL	3 x 10	Traditional	112	80%	0
Lateral Raise	3 x 10	Traditional	10	---	0

Note: Maximum Snatch = 140 kg.

When there is an increased phosphorylation, there is an increased sensitization of the actin-myosin interaction with Ca2+, which can lead to an increase in force production. From a neural perspective, it is possible that motor unit synchronization, desensitization of alpha motor neuron input, and decreased reciprocal inhibition of antagonists could result in potentiation effects.[1,18]

It is important to note that potentiation protocols appear to be impacted by the overall strength levels of the athlete and the athletes training status 1. Therefore the undulating cluster configuration should be considered an advanced set configuration technique and should only be employed with highly trained individuals. Additionally, since the major goal of the undulating cluster set is to induce a potentiation effect which may be manifested as increases in power output or velocity, it may be warranted to only include this type of set structure in the strength/power or peaking phase of a periodized training plan.

Implementing Cluster Sets into a Perioidized Training Plan

The cluster set has a large number of potential structures which allow for the matching of the set configuration to the overall goals of the phase of training.[5] Manipulations to the cluster repetition scheme, inter-repetition rest interval length, and the loading sequence can be matched to the phase of training, the goals of the training phase, and the performance characteristics of the sport being trained for.

For example, during the strength-endurance phase of a periodized training plan, the main goal for training is to induce an increased working capacity, while decreasing body fat and increasing lean body mass.[20] In this scenario the traditional set may be the most appropriate set configuration for most training exercises, especially those which are not technically challenging, such as squats or variants of the pull. Traditionally in this phase of training the use of explosive lifts such as those used by weightlifters are avoided. It is commonly believed that high repetitions schemes with technically challenging lifts such as the snatch, clean, power clean, or power snatch will result in the development of technical deficiencies which are manifested in response to repetitively performing lifts with fatigue induced technical deficiencies.[6,8] In this scenario the cluster set configuration offers a unique strategy for maintaining technical proficiency whilst developing the working capacity which is targeted by high-repetition scheme training. The employment of a short rest interval of 5–15 seconds between each repetition of the set will allow for partial recovery, allowing the athlete to maximize the quality of each repetition in the set.[5] This training strategy allows the strength and conditioning professional to utilize these lifts in the early part of the general preparation phase or the strength-endurance phase of an annual training plan.

Additionally, the use of this set configuration strategy during the strength-endurance or general preparation phase of a periodized training plan allows for the development of power-endurance. Since many sports, especially team sports such as football and rugby, require the ability to repetitively produce high power outputs[19] the capacity to maximize power-endurance would be considered an important training goal. The cluster set structure may be uniquely suited for targeting this specific performance characteristic when coupled with exercises such as the power clean or power snatch. *Table 2* gives one example of a training day contained in a strength-endurance phase. In this session, different set configuration strategies have been employed. In this example the first exercise of the session is the power clean which utilizes a standard cluster format. The cluster set is performed in a series of 5 clusters of 2 repetitions separated by 15 seconds

Table 4: Example Cluster Set Structures for the Power Snatch during a Basic Strength Phase of a Periodized Training Plan.

Type of Cluster	Sets	X	Repetitions	Example Cluster Set Repetition Loading Structure (weight / repetition)					Inter-Repetition Rest Interval (s)
Standard	1-3	X	5/1	102/1	102/1	102/1	102/1	102/1	30
	1-3	X	6/2	102/2	102/2	102/2			30
	1-3	X	5/3,2	102/3	102/2				30
Undulating	1-3	X	5/1	100/1	102/1	106/2	102/2	100/2	30
	1-3	X	6/2	102/2	106/2	100/2			30
Ascending	1-3	X	5/1	98/1	100/1	102/1	104/1	106/1	30
	1-3	X	6/2	98/2	100/2	102/2	104/2	106/2	30

Notes: 5/1 = 10 total repetitions broken into 5 clusters of 1; 6/2 = 6 total repetitions broken into 3 clusters of 2; 5/3,2 = 5 total repetitions broken into 1 cluster of 3 and 1 cluster of 2.

All weights based upon max power snatch of 120 kg (102 kg = 85% of 1 repetition maximum). Each set has an average intensity of 102 kg or 85% of 1-repetition maximum).

Table 5: Example Cluster Set Implementation during a Basic Strength Phase of Training.

Exercise	Sets x Repetitions	Set Type	Intensity kg	Intensity % 1-RM	Inter-Repetition Rest Interval(s)
Power Snatch Warm-up	3 x 10	Traditional	30	22%**	0
Clean Grip Shrugs	3 x 5	Traditional	193	133% *	0
Power Clean	3 x 5/1	Undulating Cluster			
		(130, 134, 138, 133, 130)	133	92% *	30
Clean Pull	3 x 5	Traditional	174	120% *	0
Clean Grip RDL	3 x 5	Traditional	151	104%*	0

Note: Max Power Clean = 145 kg; Max Power Snatch = 135 kg; RDL = Romanian Deadlift; * = based upon maximum power clean. A 2-3 minute rest interval is employed between each set.

for a total of 10 repetitions per set. An alternative strategy could be to use a series of 10 singles, performed with 5–10 seconds rest between each repetition or 2 series of 5 repetitions with 30 seconds rest between each series *(Table 1)*. An additional strategy which can be employed with advanced athletes, is to utilize the undulating cluster model as an alternative method for developing power-endurance *(Table 3)*. With this cluster set configuration, the loading paradigm may be structured with a series of 5 clusters of 2, in which the resistance is increased for the first 3 clusters and then unloaded with the last two clusters. The inter-repetition rest interval contained in this structure would range between 15–30 seconds depending upon the goal of phase of training and the athlete's level of fitness. By varying the number of repetitions in the series, the inter-repetitions loading pattern, and the inter-repetition rest interval during the cluster set, different physiological stressors can be introduced to the training plan which can impact different aspects of strength or power-endurance.

When shifting to the basic strength phase of a periodized training plan, the focus of training becomes the development of maximal strength[20]; thus it may be warranted to utilize traditional set structures for most non-ballistic training exercises (i.e. squats, presses, Romanian Deadlifts, etc.).[11,15] However, especially with advanced athletes the force/power generating capacity during ballistic movements can be altered with the utilization of cluster sets.[5] Overall, a greater variety of cluster set structures may be employed during this phase of training *(Table 4)*, including variations of the standard, undulating, and ascending cluster can be employed in order to maximize training effects. Additionally in this phase, it may be warranted to shift the inter-repetition rest interval closer to 30–45 seconds as the overall intensity of the set is substantially higher, which would require longer inter-repetition rest intervals in order to facilitate recovery.[5]

In this phase the undulating cluster is often employed as it allows the athlete to begin using higher intensities in training.[5] When planning and implementing an undulating cluster the overall intensity of the set is defined by the average kilograms lifted across the set. For example in *Table 4* an example undulating cluster is presented where the average intensity is 102 kg, or 85% of the athletes maximal power snatch. In order to achieve the 102 kg loading average for the planned cluster set, the athlete might perform a series of singles at 100, 102, 106, 102, and 100kg for each set with 30 seconds between each repetition. This basic structure can then be incorporated into a training session with ballistic or explosive exercises, while traditional sets are utilized for the other exercises contained in the session *(Table 5)*. In this session a couple of advanced training techniques are employed. Firstly, two strategies have been employed in order to create a potentiation effect. The first potentiation strategy involves the performance of the clean grip shoulder shrug prior to the power clean. Theoretically, the ordering of the exercise in this fashion should allow for an increase in force production, movement velocity, and power output during the sets of power cleans. The second potentiation strategy involves structuring the power clean sets as undulating clusters.[5] In this format the unloading portion of the set will result in significantly higher movement velocities, power outputs, and barbell displacements.[6] While these strategies have the potential to be very effective in elevating strength levels, they are probably best suited for advanced athletes who already have high levels of strength[1] and tolerate the additional training stress introduced when using cluster sets.

As the training plan shifts from the basic-strength into the strength-power phase of the periodized training plan, the focus of training centers on further increases in maximal strength while also elevating power generating capacity.[14,20] Due to its ability to maintain power output across the training set, the cluster set

Table 6: Example Ascending Set Structures.

Type of Cluster	Set (#)	Set	X	Repetitions	Average Intensity	%1 RM	Example Cluster Set Repetition Loading Structure (weight / repetition)			Inter-Repetition Rest Interval (s)
Ascending	1	1	X	3/1	102	85%	98/1	102/1	106/1	30
	2	1	X	3/1	106	88%	102/1	106/1	110/1	30
	3	1	X	3/1	110	91%	106/1	110/1	114/1	30

Notes: 3/1 = 3 total repetitions broken into 3 clusters of 1; RM = Repetition Maximum.
All weights based upon max power snatch of 120 kg (102 kg = 85% of 1 repetition maximum).

Table 7: Example Cluster Set Implementation during A Strength-Power Phase of Training.

Exercise	Sets x Repetitions	Set Type	Intensity		Inter-Repetition Rest Interval(s)
			kg	% 1-RM	
Speed Squats	3 x 3	Traditional	100	50%	0
Power Clean	1 x 3/1	Ascending Cluster	115, 120, 125	86%	30
	1 x 3/1	Ascending Cluster	120, 125, 130	89%	30
	1 x 3/1	Ascending Cluster	125, 130, 135	93%	30
Push Jerk*	3 x 3	Traditional	120	89%	0

Note: Max Power Clean = 140 kg; Max Back Squat = 200 kg; Max push jerk = 135 kg. A 2-3 minute rest interval is employed between each set.
* = front squat first repetition of each set.

appears to be ideally suited for this phase of the training plan.[5] In this phase it is likely that the undulating and ascending cluster sets are the predominantly used set configurations, as they both have potential to significantly impact power generating capacity. In this phase since the repetition scheme is decreased, 20 greater inter-repetition loading paradigms can be employed, thus allowing for greater stimuli for post-activation potentiation to be created. Since intensity of the repetitions is very high with these set configurations the rest interval should be somewhere between 30–45 seconds in order to allow the athlete enough recovery time in order to maximize power out.

In the attempt to elevate maximal force generating capacity the strength and conditioning professional should consider the use of an ascending cluster set structure during the strength-power phase. The ascending cluster set would have increasing intensities with each repetition of the set and the average intensity of each set would also increase *(Table 6)*. For example, the average intensity of the three sets may be performed at 85%, 88%, and 91% of 1-RM. Additionally, the intensity of each repetition contained in the cluster set will increase. Therefore if set 1 is performed with an average intensity of 102 kg or 85% of 1-RM, the athlete would lift 98, 102, 106 kg across the cluster set and utilize rest intervals between 30–45 seconds. Overall, this method of cluster set loading is best suited for elevating peak force generating capacity and may be best suited for partial movements or power movements, such as the power clean or power snatch.[5] This type of loading paradigm appears to result in a large amount of fatigue, which manifests itself in response to using 3 ascending sets in a row. This increased potential for developing fatigue needs to be considered in the context of the overall training plan and addressed when constructing the individual training sessions. *Table 7* gives an example of a training session which utilizes ascending cluster sets. In this example the ascending cluster set structure is utilized with 30 second inter-repetition rest intervals for the power clean, while traditional sets are used for all other exercises. Because of the potential to develop high levels of fatigue when using ascending cluster sets it is important to carefully craft the training regime. Additionally, the implementation of a recovery intervention program, which utilizes activities such as massage, contrast baths, or ice therapies, may be warranted to aid in the athletes ability to tolerate this type of intense loading.[12,21,22]

Conclusions

Based upon the available scientific data it appears that the different variants of the cluster set allow for unique methods of introducing variation into the periodized training plan. The ability to modify the cluster set structure allows the strength and conditioning professional the ability to target specific physiological and performance characteristics. Specifically, this unique tool should be considered when working with strength-power athletes or athletes who are utilizing power based exercises in their preparation. While there is scientific support for the utilization of cluster sets, significantly more scientific research must be conducted in order to understand the performance and physiological effects of various cluster set models. Additionally, more research is needed to understand the optimization of program considerations when employing cluster sets.

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